

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Dial indicator assembly, Part Number 3376050
- Magnetic base indicator holder, Part Number 3377399

Additional Service Items

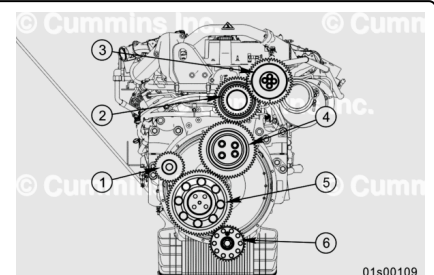
- No additional service items required.

General Information

Two types of Rear Gear Trains (RGTs) are available, with spur gears or helical gears.

RGT

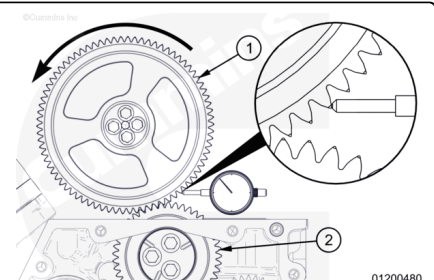
1. Air compressor gear
2. Compound camshaft idler gear
3. Camshaft gear
4. Camshaft gear idler gear
5. Crankshaft idler gear
6. Crankshaft gear.



Measure

This procedure is to measure gear backlash. Do **not** allow the adjacent mating gear to turn during the measurement.

Rotate the gear being measured (1) **clockwise** to remove any backlash with the mating gear (2).



Position dial indicator assembly, Part Number 3376050, with magnetic base indicator holder, Part Number 3377399, so the tip is contacting the surface of the gear tooth, as shown in the illustration. Do **not** allow the mating gear (2) to turn.

Set the dial indicator to zero.

Rotate the gear (1) **counterclockwise**. Read the dial indicator.

Measure the backlash at the following locations:

- Rear camshaft gear to rear camshaft idler gear
- Crankshaft gear (rear) to rear crankshaft idler gear
- Crankshaft idler gear to camshaft idler gear
- Air compressor gear to crankshaft idler gear
- Rear Engine Power Takeoff (REPTO) gear to crankshaft idler gear.

If gear backlash is **not** within specifications, the gear is out of specification and **must** be replaced.

With Spur Gear

Camshaft Gear to Compound Camshaft Idler Gear Backlash

mm		in
0.02	MIN	0.001
0.22	MAX	0.009

Camshaft Gear Idler Gear to Compound Camshaft Idler Gear Backlash

mm		in
0.02	MIN	0.001
0.21	MAX	0.008

Camshaft Gear Idler Gear to Crankshaft Idler Gear Backlash

mm		in
0.05	MIN	0.002
0.286	MAX	0.011

Crankshaft Idler Gear to Air Compressor Gear Backlash

mm		in
0.05	MIN	0.002
0.234	MAX	0.009

Crankshaft Idler Gear to Crankshaft Gear Backlash		
mm		in
0.08	MIN	0.003
0.28	MAX	0.011

With Helical Gear

Camshaft Gear to Compound Camshaft Idler Gear Backlash		
mm		in
0.02	MIN	0.001
0.15	MAX	0.006

Camshaft Gear Idler Gear to Compound Camshaft Idler Gear Backlash		
mm		in
0.02	MIN	0.001
0.15	MAX	0.006

Camshaft Gear Idler Gear to Crankshaft Idler Gear Backlash		
mm		in
0.03	MIN	0.001
0.14	MAX	0.006

Crankshaft Idler Gear to Air Compressor Gear Backlash		
mm		in
0.03	MIN	0.001
0.15	MAX	0.006

Crankshaft Idler Gear to Crankshaft Gear Backlash		
mm		in
0.03	MIN	0.001
0.15	MAX	0.006